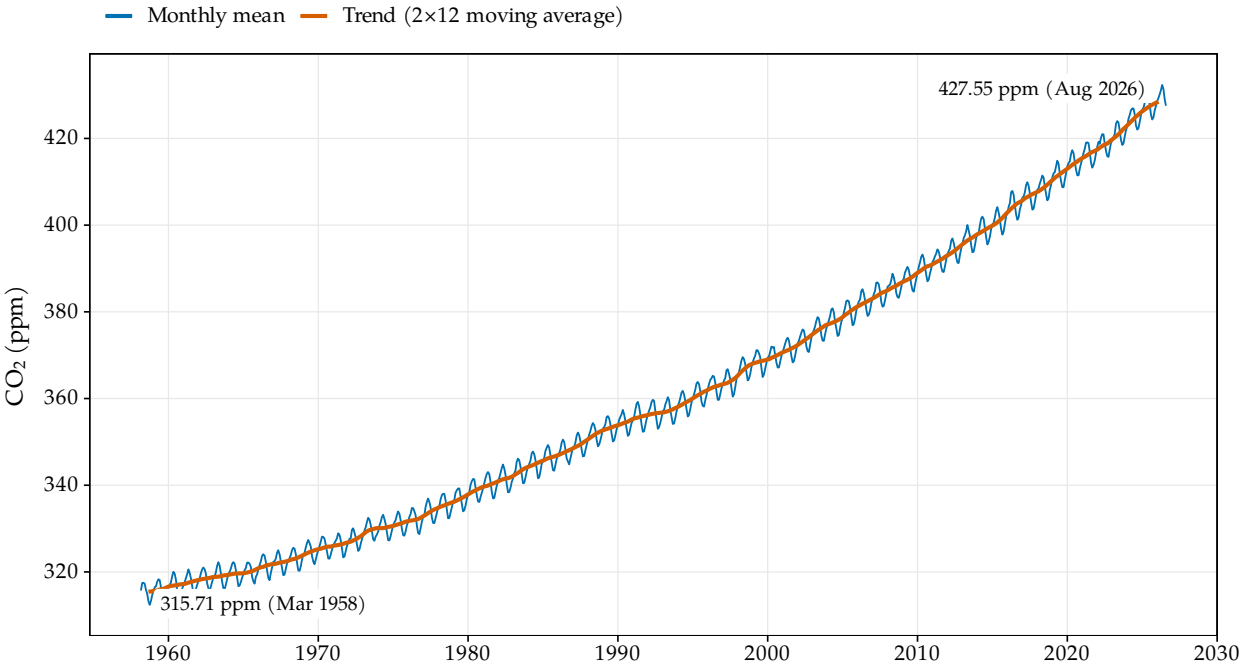


Figure 1. Monthly mean atmospheric CO_2 at Mauna Loa, March 1958 to August 2026, with the 2×12 moving-average trend.

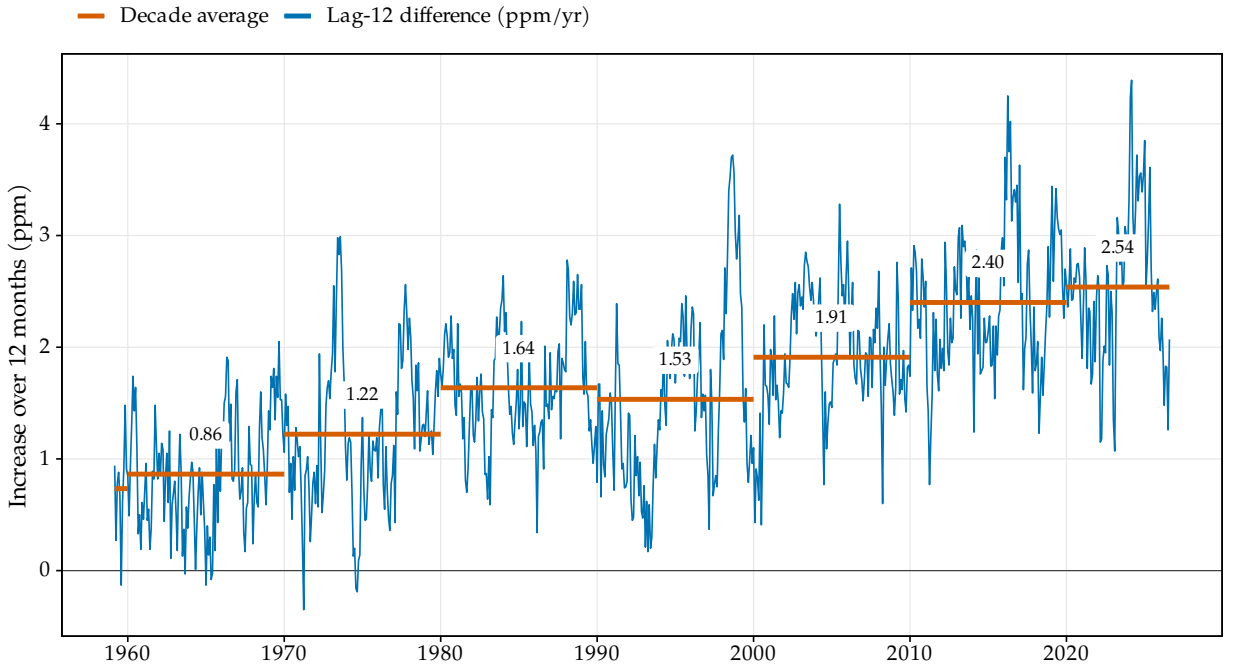


Figure 2. Twelve-month change $\nabla_{12} X_t$ (ppm per year) with decade averages: growth tripled from the 1960s to the 2020s.

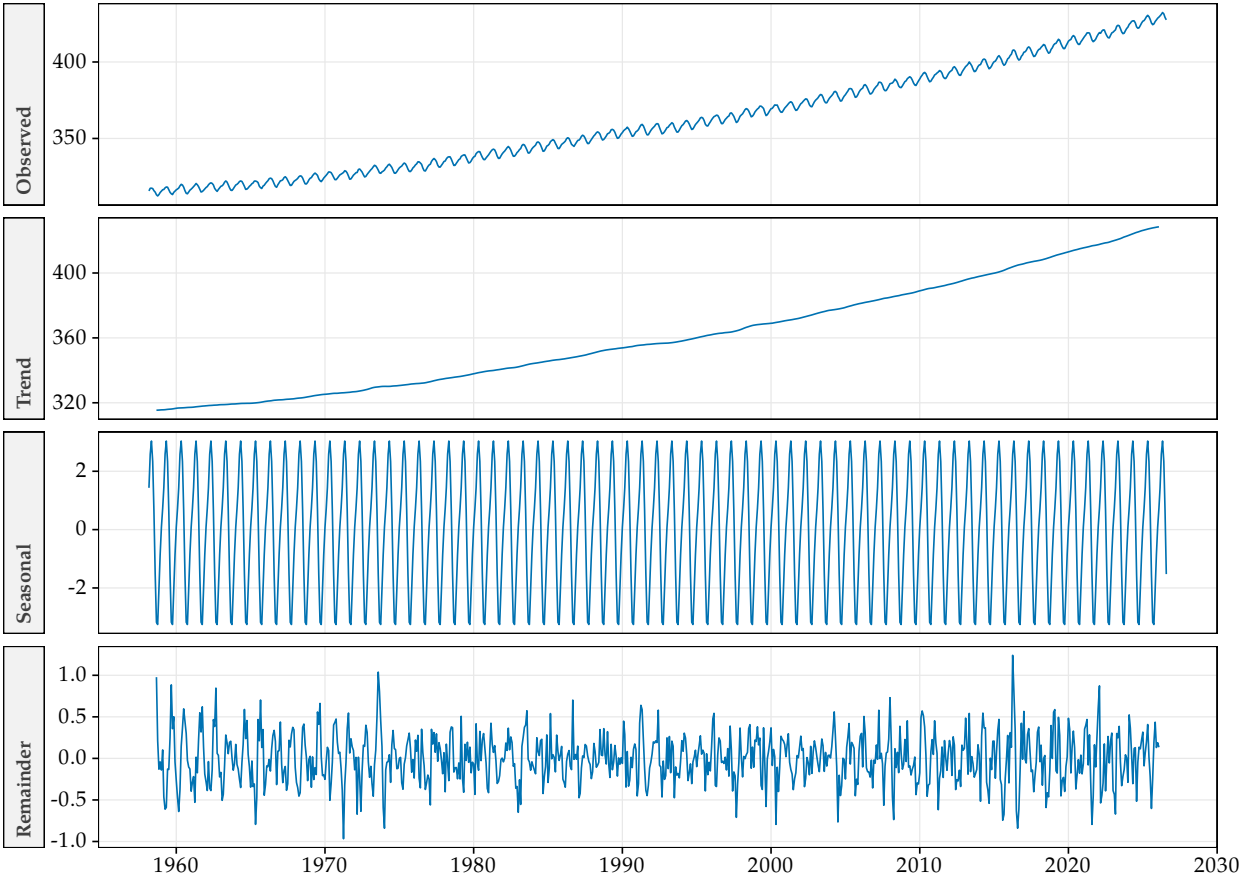


Figure 3. Classical additive decomposition of X_t into trend, seasonal and remainder components.

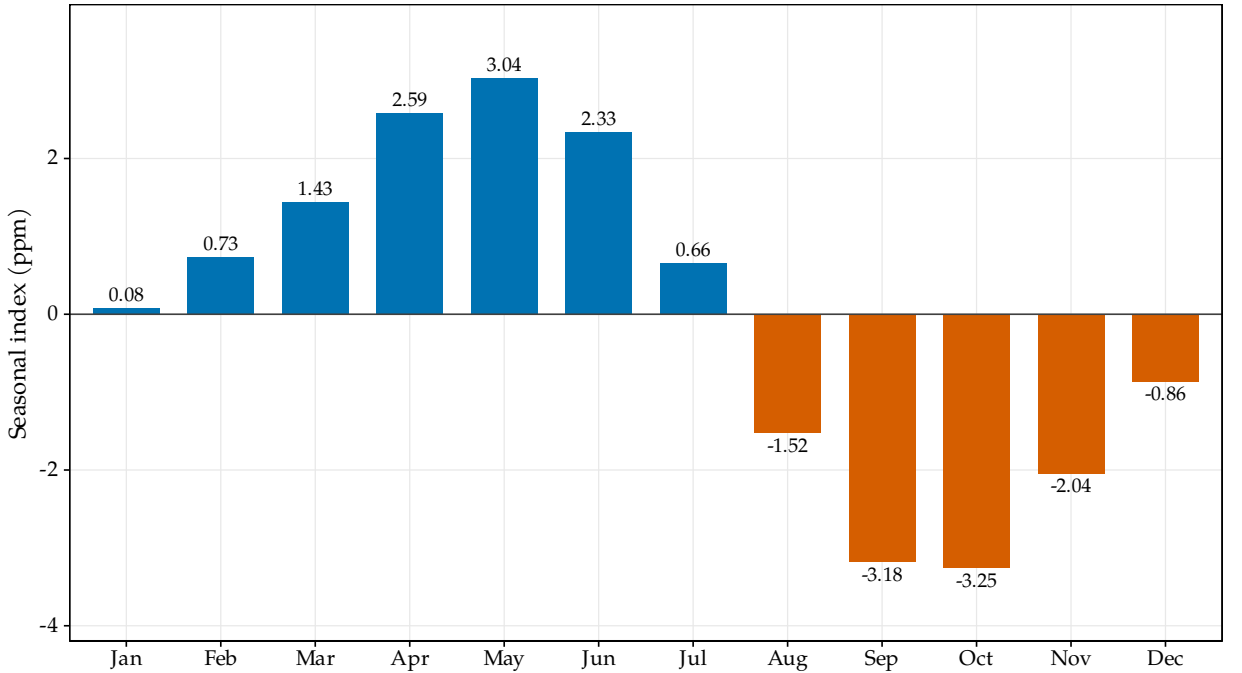


Figure 4. Monthly seasonal indices from the additive decomposition: peak in May (+3.04 ppm), trough in October (−3.25 ppm).

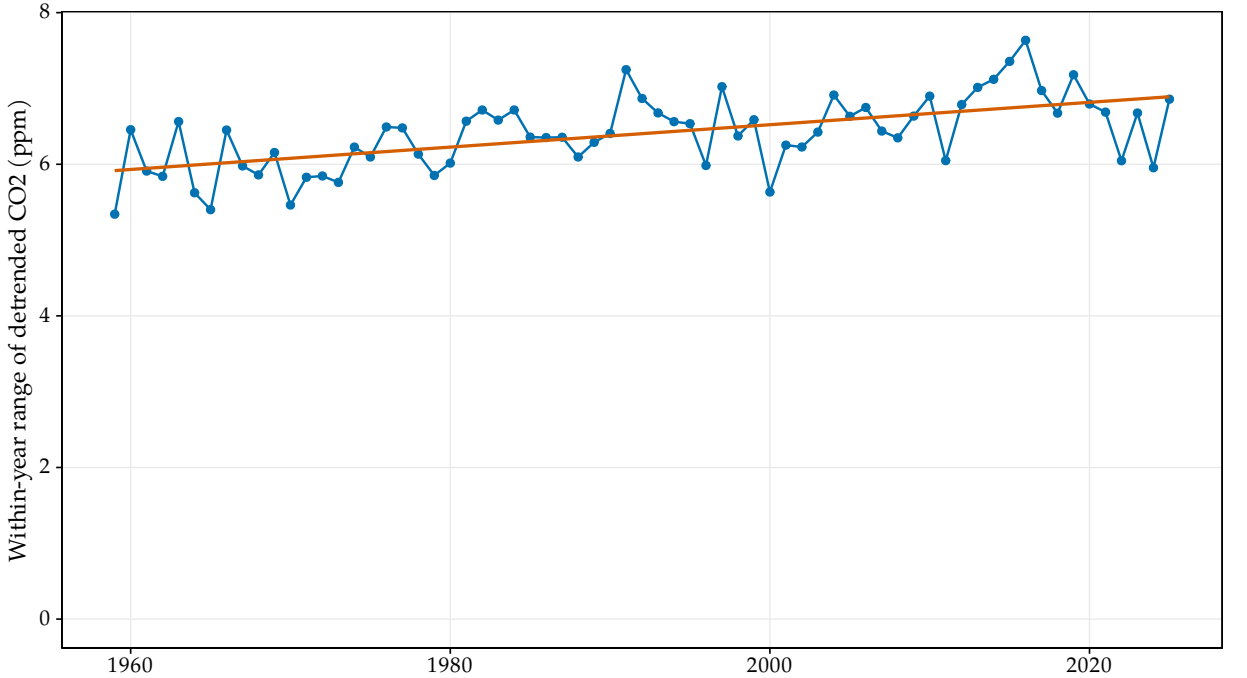


Figure 5. Within-year range of the detrended series by year, with a linear trend. The amplitude does not grow in proportion to the level, so no transformation is needed.

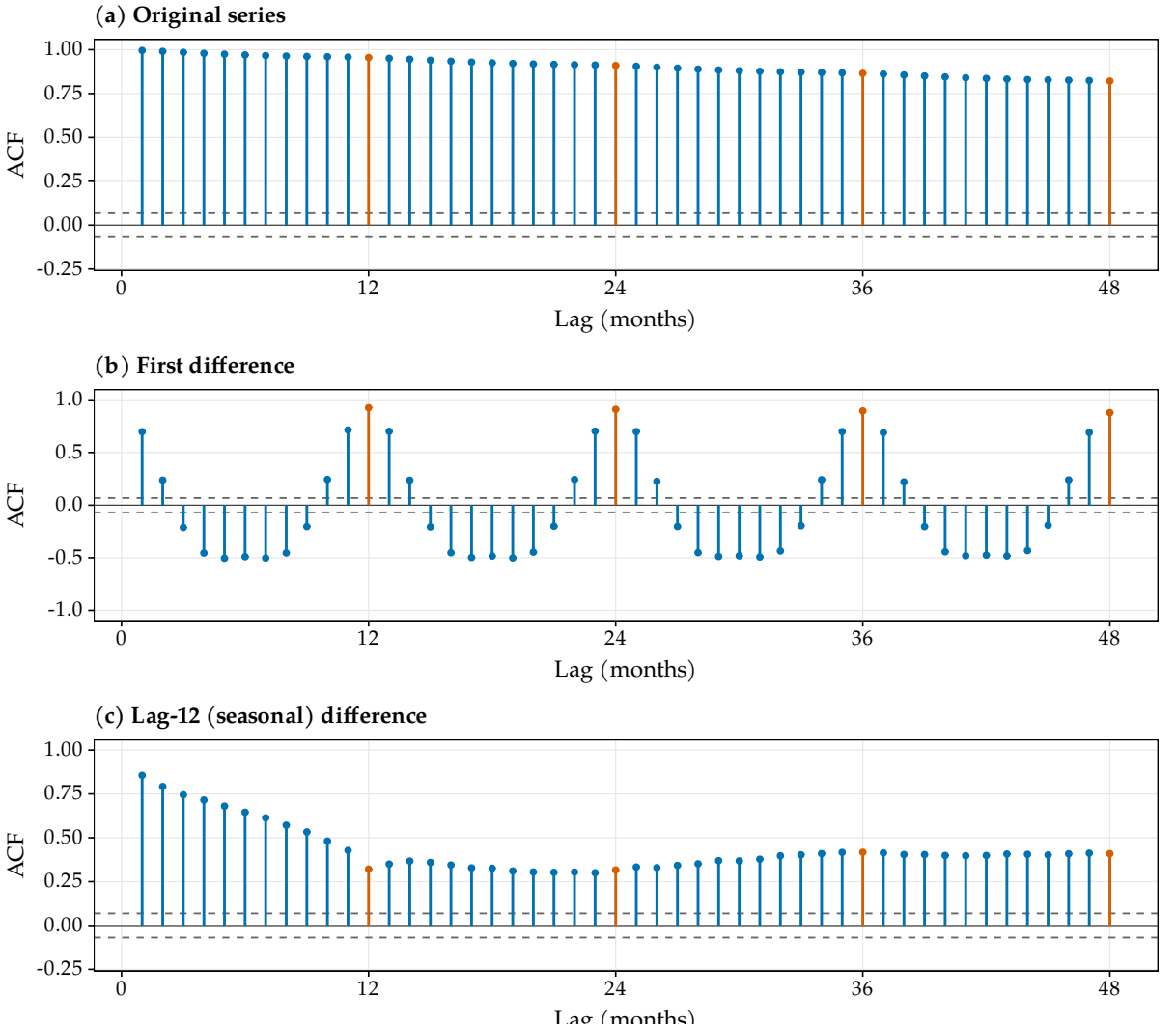


Figure 6. Sample ACF of (a) X_t, (b) ∇X_t and (c) $\nabla_{12} X_t$. Seasonal lags are highlighted; dashed lines are the $\pm 1.96/\sqrt{n}$ bounds.

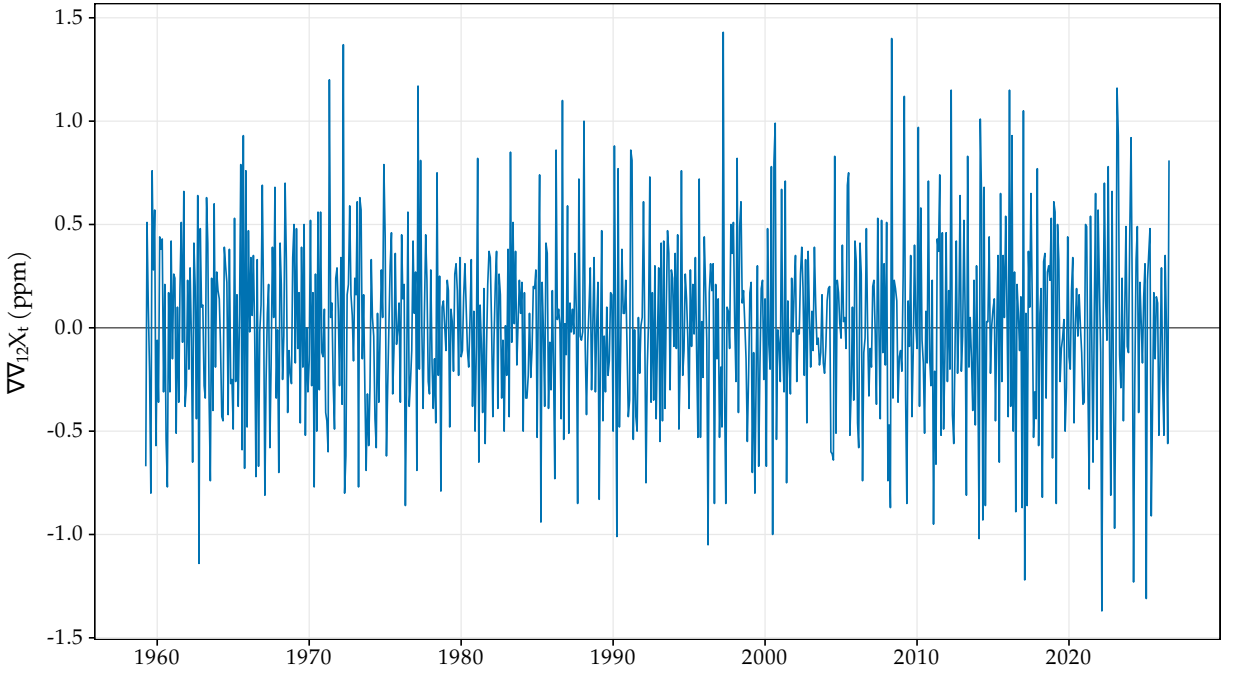


Figure 7. The stationary series $W_t = \nabla\nabla_{12}X_t$ used for model identification.

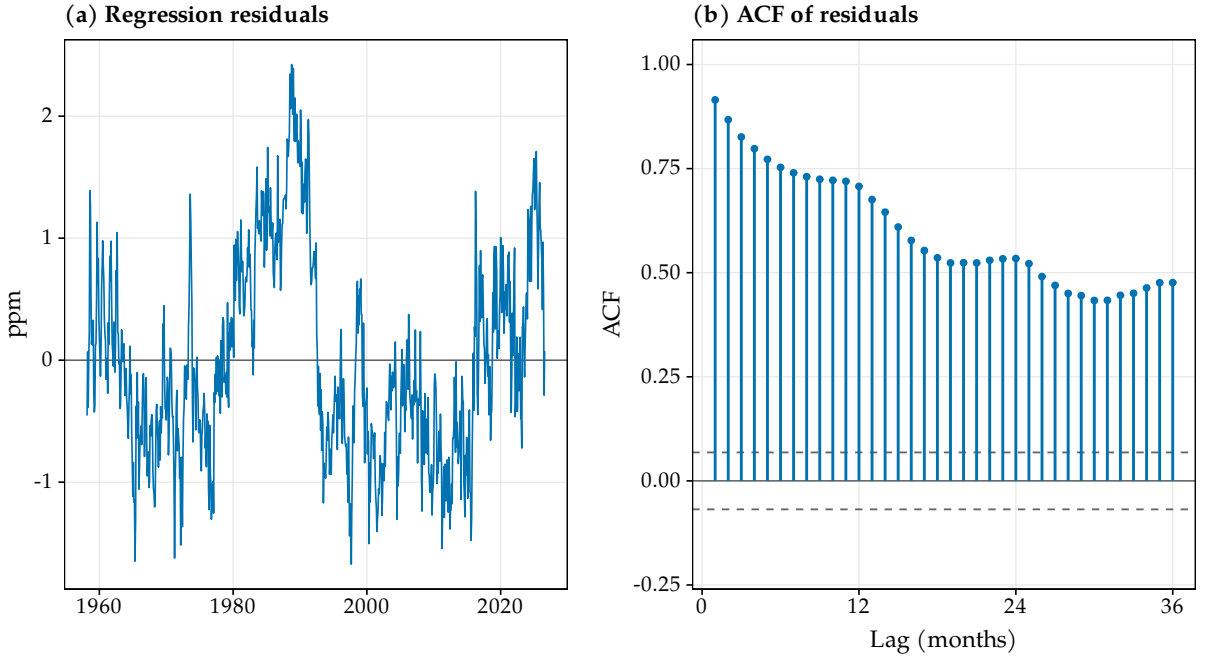


Figure 8. Regression on a quadratic trend and monthly dummies: (a) residuals and (b) their sample ACF. The residuals persist for years.

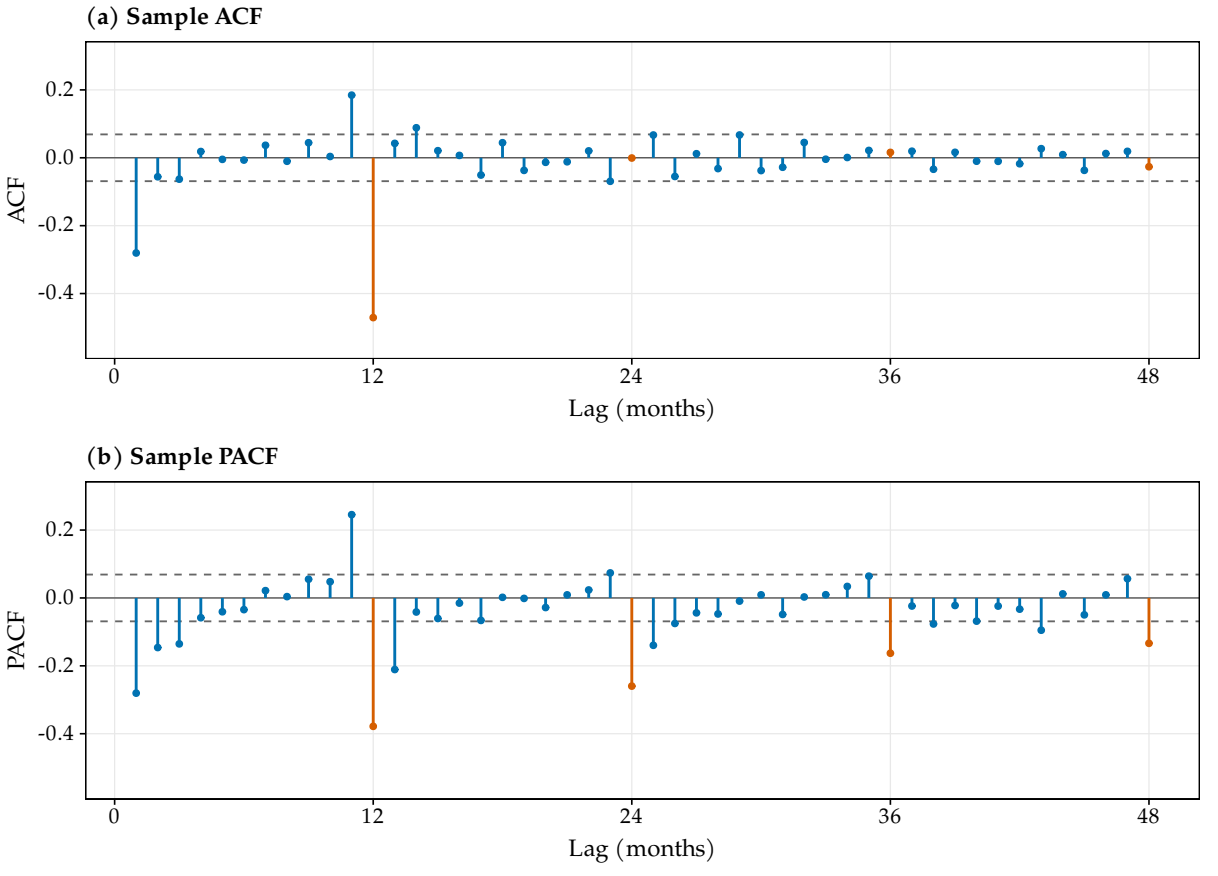


Figure 9. Sample ACF (a) and PACF (b) of W_t. The ACF cuts off after lags 1 and 12; the PACF tails off.

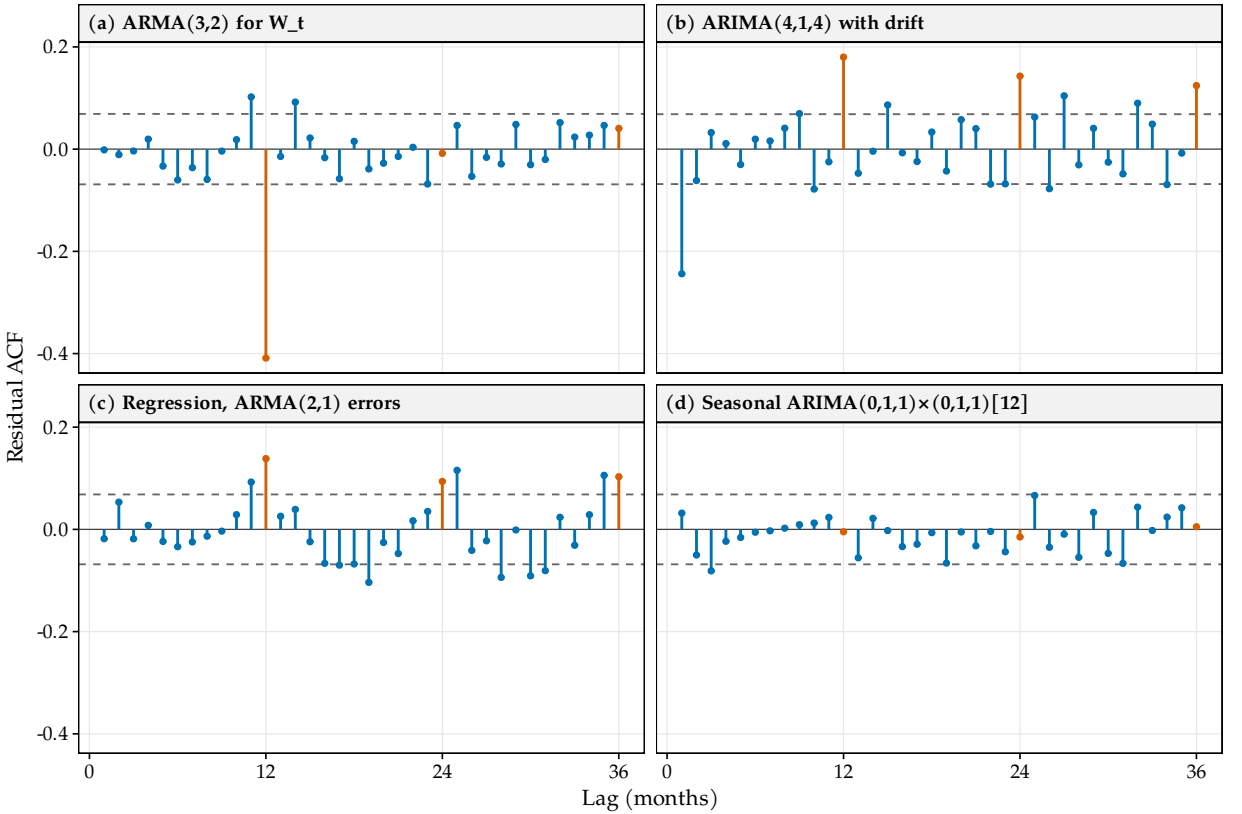


Figure 10. Residual ACF of the best model in each of the four families. Only the seasonal ARIMA (d) has no spikes at the seasonal lags.

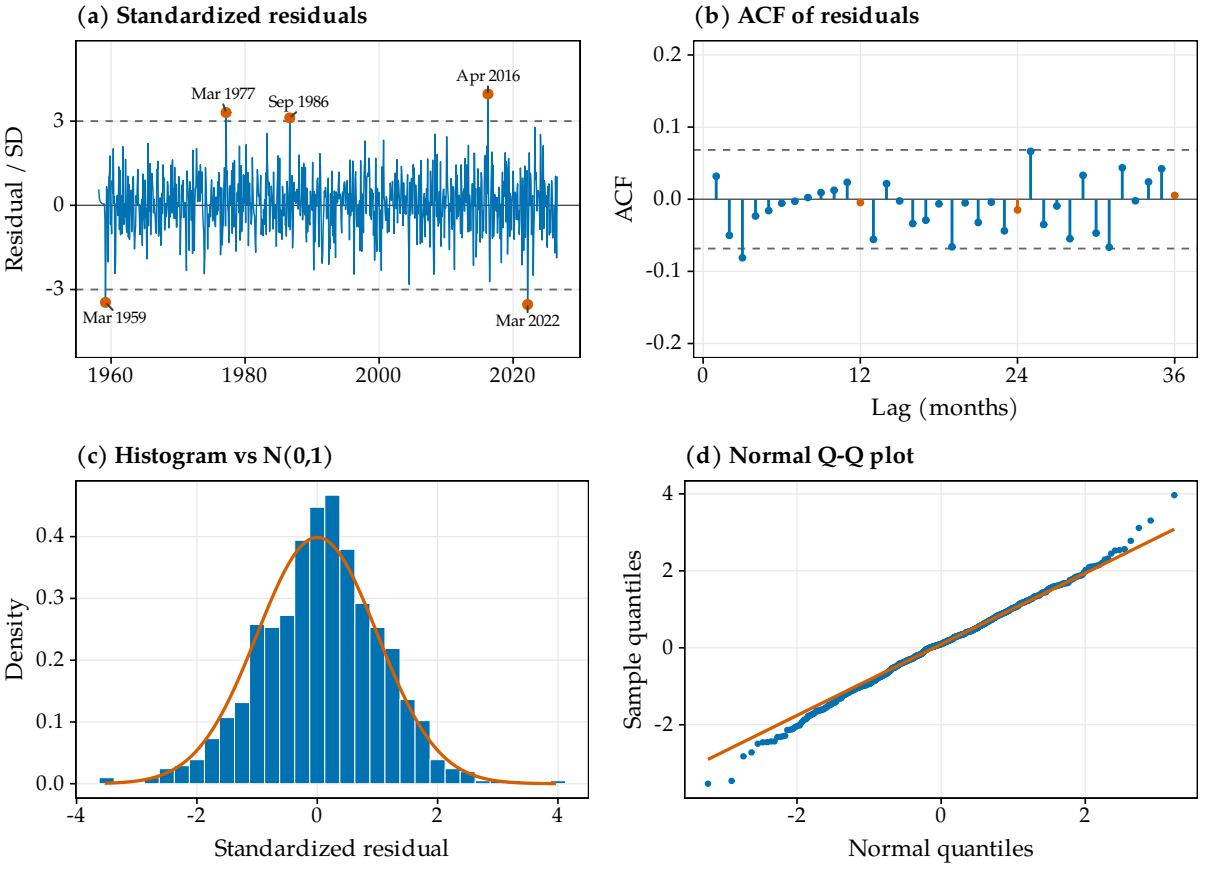


Figure 11. Residual diagnostics for ARIMA$(0,1,1) \times (0,1,1)_{12}$: (a) standardized residuals with the five months beyond ± 3 labeled, (b) residual ACF, (c) histogram with the $N(0,1)$ density, (d) normal Q–Q plot.

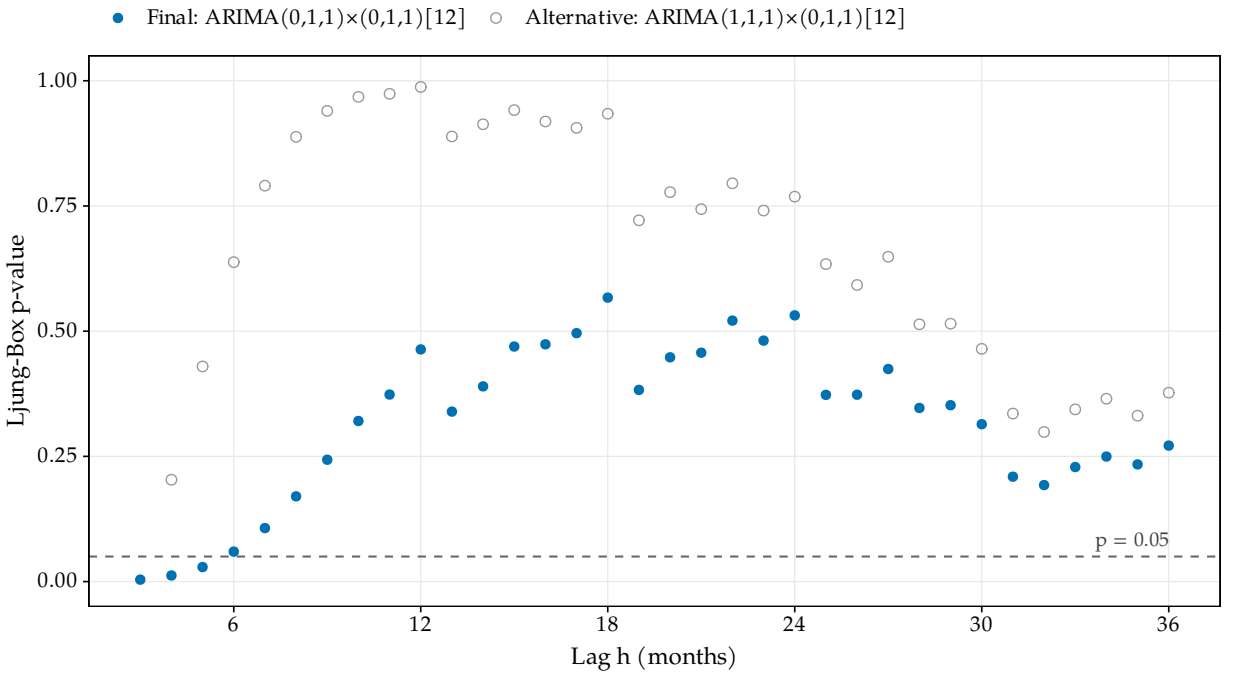


Figure 12. Ljung–Box p-values by lag for the final and the alternative model; the dashed line is $p = 0.05$.

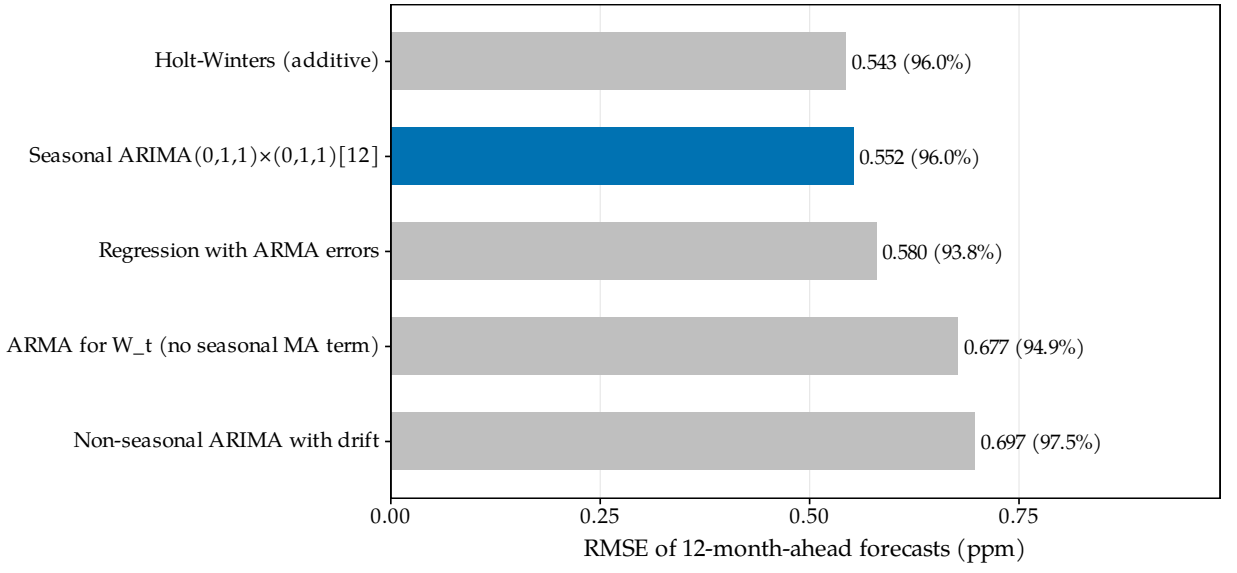


Figure 13. Rolling-origin evaluation: RMSE of 12-month-ahead forecasts from 23 origins (276 forecasts per method), with 95% interval coverage in parentheses.

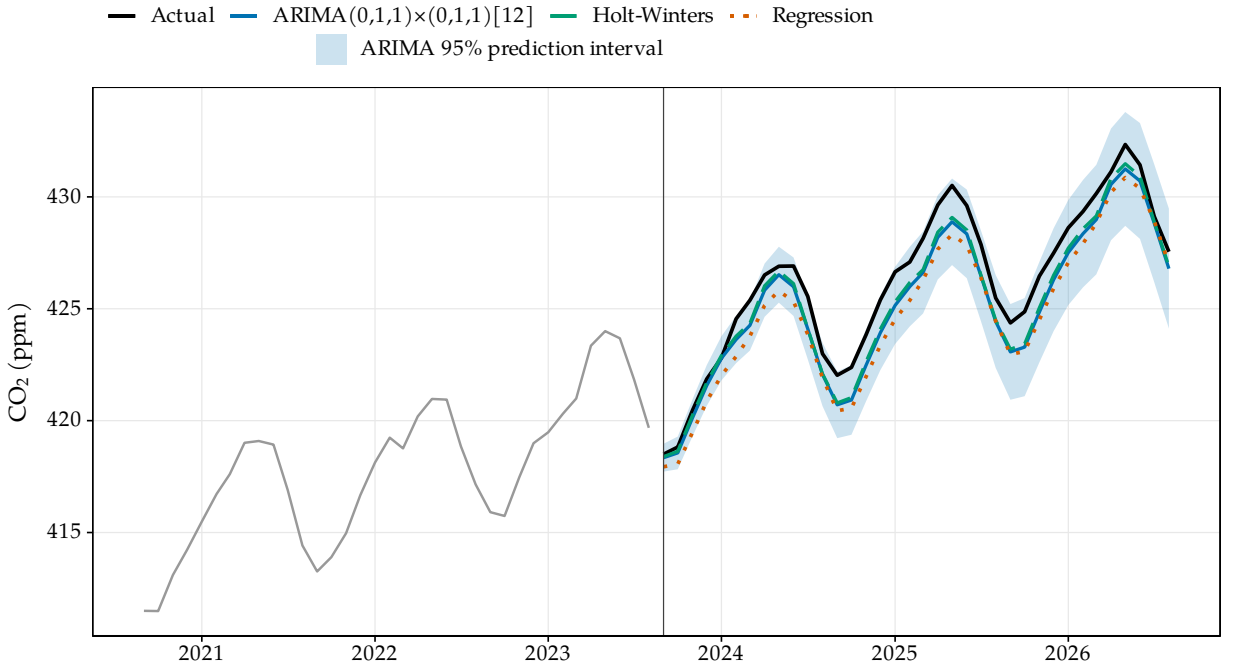


Figure 14. Hold-out forecasts for September 2023 to August 2026 from models fitted through August 2023, with the ARIMA 95% prediction interval.

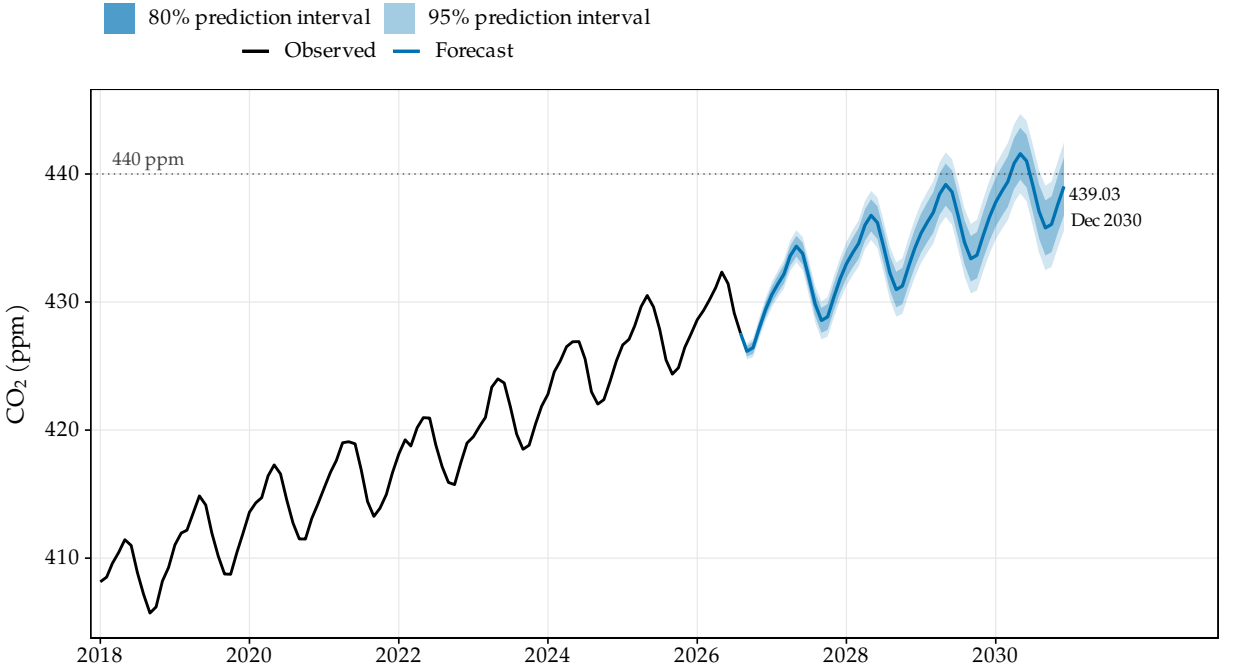


Figure 15. Forecast of monthly CO_2 to December 2030 from ARIMA$(0,1,1) \times (0,1,1)_{12}$, with 80% and 95% prediction intervals.

Table 1. Descriptive statistics of the monthly CO_2 series.

Statistic	Value
Observations (months)	822
First month	Mar 1958
Last month	Aug 2026
Mean (ppm)	361.36
Standard deviation (ppm)	33.46
Minimum (ppm)	312.42 (Oct 1958)
Maximum (ppm)	432.34 (May 2026)
Average annual increase (ppm/yr)	1.68

Table 2. Mean level, mean annual increase and seasonal amplitude by decade.

Decade	Mean CO_2 (ppm)	Annual increase (ppm/yr)	Seasonal amplitude (ppm)
1950s	315.64	0.73	5.34
1960s	320.29	0.86	6.02
1970s	330.86	1.22	6.02
1980s	345.65	1.64	6.41
1990s	360.58	1.53	6.63
2000s	378.77	1.91	6.43
2010s	400.41	2.40	6.97
2020s	421.32	2.54	6.50

Table 3. Monthly seasonal indices from the additive decomposition (ppm).

Month	Seasonal index (ppm)
Jan	0.083
Feb	0.729
Mar	1.433
Apr	2.587
May	3.036
Jun	2.334
Jul	0.656
Aug	−1.525
Sep	−3.177
Oct	−3.253
Nov	−2.043
Dec	−0.861

Table 4. Augmented Dickey–Fuller tests (H_0: unit root). Seasonally adjusted series at lags 0–2; reject H_0 when τ is below the critical value.

Series	Type	Lags	τ	5% critical value	1% critical value	Decision
Seasonally adjusted X_t	Single mean	0	3.57	−2.86	−3.43	Fail to reject (unit root)
Seasonally adjusted X_t	Single mean	1	4.53	−2.86	−3.43	Fail to reject (unit root)
Seasonally adjusted X_t	Single mean	2	4.85	−2.86	−3.43	Fail to reject (unit root)
Seasonally adjusted X_t	Trend	0	−0.88	−3.41	−3.96	Fail to reject (unit root)
Seasonally adjusted X_t	Trend	1	−0.79	−3.41	−3.96	Fail to reject (unit root)
Seasonally adjusted X_t	Trend	2	−0.62	−3.41	−3.96	Fail to reject (unit root)
X_t (13 augmentation lags)	Trend	13	−0.33	−3.41	−3.96	Fail to reject (unit root)
Lag-12 difference	Single mean	0	−7.91	−2.86	−3.43	Reject H_0 (stationary)
Lag-12 difference	Single mean	1	−6.14	−2.86	−3.43	Reject H_0 (stationary)
Lag-12 difference	Single mean	2	−5.37	−2.86	−3.43	Reject H_0 (stationary)
Regular + seasonal difference	Single mean	0	−37.89	−2.86	−3.43	Reject H_0 (stationary)
Regular + seasonal difference	Single mean	1	−26.26	−2.86	−3.43	Reject H_0 (stationary)
Regular + seasonal difference	Single mean	2	−22.05	−2.86	−3.43	Reject H_0 (stationary)

Table 5. Regression of X_t on a quadratic trend and monthly dummies (estimation approach), with R^2 and the Durbin–Watson statistic.

Term	Estimate	SE	t	p
Intercept	314.7597	0.1283	2,453.05	< 0.001
t	0.0621	0.0005	128.89	< 0.001
t^2	0.0001	0.0000	165.35	< 0.001
Month: Feb	0.6444	0.1403	4.59	< 0.001
Month: Mar	1.3417	0.1398	9.60	< 0.001
Month: Apr	2.4979	0.1398	17.87	< 0.001
Month: May	2.9496	0.1398	21.10	< 0.001
Month: Jun	2.2487	0.1398	16.09	< 0.001
Month: Jul	0.5630	0.1398	4.03	< 0.001
Month: Aug	−1.5948	0.1398	−11.41	< 0.001
Month: Sep	−3.2629	0.1403	−23.26	< 0.001
Month: Oct	−3.3378	0.1403	−23.80	< 0.001
Month: Nov	−2.1275	0.1403	−15.17	< 0.001
Month: Dec	−0.9436	0.1403	−6.73	< 0.001
R^2	0.9994			
Durbin-Watson D	0.1693			

Table 6. Ljung–Box tests for $W_t = \nabla\nabla_{12}X_t$ (H_0: white noise).

Lag	Q	p
6	69.99	< 0.001
12	283.32	< 0.001
18	295.43	< 0.001
24	301.19	< 0.001

Table 7. Candidate seasonal ARIMA models estimated by maximum likelihood, sorted by SBC.

Model	Parameters	Log-likelihood	AIC	SBC	ΔSBC	Ljung–Box $Q(24)$ p
ARIMA$(0,1,1)\times(0,1,1)_{12}$	2	−222.65	451.30	465.39	0.00	0.532
ARIMA$(1,1,1)\times(0,1,1)_{12}$	3	−219.79	447.58	466.37	0.98	0.769
ARIMA$(0,1,2)\times(0,1,1)_{12}$	3	−220.30	448.59	467.38	1.99	0.723
ARIMA$(0,1,1)\times(0,1,2)_{12}$	3	−222.65	453.30	472.08	6.69	0.472
ARIMA$(0,1,1)\times(1,1,1)_{12}$	3	−222.65	453.30	472.08	6.69	0.472
ARIMA$(1,1,1)\times(1,1,1)_{12}$	4	−219.79	449.57	473.05	7.66	0.715
ARIMA$(2,1,0)\times(0,1,1)_{12}$	3	−226.32	460.64	479.42	14.03	0.213
ARIMA$(1,1,0)\times(0,1,1)_{12}$	2	−233.41	472.83	486.92	21.53	0.038
ARIMA$(0,1,1)\times(1,1,0)_{12}$	2	−331.67	669.34	683.43	218.04	<0.001

Table 8. Best model in each of the four families (k = number of ARMA parameters): residual checks and 36-month hold-out accuracy.

				Ljung–Box p		Hold-out	
Family	Best model	k	ACF lag 12	$Q(12)$	$Q(24)$	RMSE	Coverage (%)
ARMA	ARMA(3,2) for W_t	5	-0.409	< 0.001	< 0.001	0.655	100.0
ARIMA	ARIMA(4,1,4) with drift	8	0.180	< 0.001	< 0.001	2.127	100.0
Regression with ARMA errors	Trend + dummies, ARMA(2,1) errors	3	0.138	< 0.001	< 0.001	1.207	55.6
Seasonal ARIMA	ARIMA$(0,1,1) \times (0,1,1)_{12}$	2	-0.005	0.464	0.532	1.110	97.2

Table 9. Regression with ARMA errors: the AR coefficients sum to nearly 1 and the residuals fail the Ljung–Box test.

Error model	Sum of AR coefficients	Ljung–Box $Q(12)$ p	Ljung–Box $Q(24)$ p
ARMA(1,0)	0.914	< 0.001	< 0.001
ARMA(2,0)	0.930	< 0.001	< 0.001
ARMA(2,1)	0.990	< 0.001	< 0.001

Table 10. Parameter estimates, z-tests and 95% confidence intervals (maximum likelihood and conditional least squares).

Model	Method	Parameter	Estimate	SE	z	p	95% CI
Final: ARIMA$(0,1,1)\times(0,1,1)_{12}$	ML	θ_1 (MA(1))	0.3885	0.0367	10.59	<0.001	[0.317, 0.460]
Final: ARIMA$(0,1,1)\times(0,1,1)_{12}$	ML	Θ_1 (seasonal MA(1))	0.8700	0.0170	51.28	<0.001	[0.837, 0.903]
Final: ARIMA$(0,1,1)\times(0,1,1)_{12}$	CLS	θ_1 (MA(1))	0.4172	0.0360	11.60	<0.001	[0.347, 0.488]
Final: ARIMA$(0,1,1)\times(0,1,1)_{12}$	CLS	Θ_1 (seasonal MA(1))	0.8116	0.0194	41.84	<0.001	[0.774, 0.850]
Alternative: ARIMA$(1,1,1)\times(0,1,1)_{12}$	ML	ϕ_1 (AR(1))	0.2151	0.0837	2.57	0.010	[0.051, 0.379]
Alternative: ARIMA$(1,1,1)\times(0,1,1)_{12}$	ML	θ_1 (MA(1))	0.5744	0.0701	8.20	<0.001	[0.437, 0.712]
Alternative: ARIMA$(1,1,1)\times(0,1,1)_{12}$	ML	Θ_1 (seasonal MA(1))	0.8699	0.0169	51.50	<0.001	[0.837, 0.903]

Table 11. Residual diagnostics for the final model, the final model with the five outlying months set to missing, and the alternative ARIMA$(1,1,1) \times (0,1,1)_{12}$.

	Final model		Outlying months missing		Alternative model	
Test	Statistic	p	Statistic	p	Statistic	p
Ljung–Box $Q(6)$	9.050	0.060	7.985	0.092	1.696	0.638
Ljung–Box $Q(12)$	9.740	0.464	10.306	0.414	2.213	0.988
Ljung–Box $Q(18)$	14.427	0.567	14.871	0.534	7.730	0.934
Ljung–Box $Q(24)$	20.825	0.532	18.732	0.662	16.017	0.769
Ljung–Box $Q(12)$, squared	12.398	0.414	16.935	0.152	9.987	0.617
Ljung–Box $Q(24)$, squared	35.359	0.063	40.293	0.020	32.456	0.116
Shapiro–Wilk	0.996	0.048	0.997	0.211	0.996	0.037
Anderson–Darling	0.800	0.038	0.630	0.101	0.771	0.045
Jarque–Bera	10.306	0.006	4.735	0.094	11.132	0.004

Table 12. Months with standardized residuals beyond ± 3.

Month	Observed (ppm)	Standardized residual
Mar 1959	316.65	−3.46
Mar 1977	334.96	3.31
Sep 1986	345.48	3.11
Apr 2016	407.60	3.97
Mar 2022	418.76	−3.53

Table 13. Sensitivity of the estimates to the five outlying months.

Data	θ_1	Θ_1	$\hat{\sigma}_Z^2$
All months	0.3885	0.8700	0.1012
Outlying months set to missing	0.3838	0.8689	0.0978

Table 14. Rolling-origin accuracy of 12-month-ahead forecasts (23 origins, August 2000–2022).

Method	RMSE	MAE	95% PI coverage (%)	Forecasts
Holt–Winters (additive)	0.543	0.403	96.0	276
Seasonal ARIMA$(0,1,1)\times(0,1,1)_{12}$	0.552	0.411	96.0	276
Regression with ARMA errors	0.580	0.445	93.8	276
ARMA for W_t (no seasonal MA term)	0.677	0.521	94.9	276
Non-seasonal ARIMA with drift	0.697	0.543	97.5	276

Table 15. Holt–Winters smoothing parameters, full sample.

Parameter	Value
α (level)	0.5337
β (trend)	0.0139
γ (seasonal)	0.2990
Final level (ppm)	429.1464
Final slope (ppm/month)	0.2030

Table 16. Hold-out forecast accuracy, September 2023 to August 2026.

Method	RMSE	MAE	MAPE (%)	95% PI coverage (%)
Final: ARIMA$(0,1,1) \times (0,1,1)_{12}$	1.110	1.010	0.237	97.2
ARMA for W_t (no seasonal MA term)	0.655	0.501	0.117	100.0
Non-seasonal ARIMA with drift	2.127	1.930	0.452	100.0
Regression with ARMA errors	1.207	1.113	0.261	55.6
Alternative: ARIMA$(1,1,1) \times (0,1,1)_{12}$	1.168	1.072	0.251	83.3
Holt–Winters (additive)	0.981	0.875	0.205	97.2
Regression: quadratic trend + seasonal dummies	1.491	1.413	0.331	52.8
Seasonal naive	6.448	6.069	1.422	25.0

Table 17. Forecasts of monthly CO_2 (ppm) with 80% and 95% prediction intervals and the Holt–Winters forecast.

Month	Forecast	80% PI	95% PI	Holt–Winters
Sep 2026	426.16	[425.75, 426.56]	[425.53, 426.78]	426.19
Dec 2026	429.39	[428.80, 429.99]	[428.49, 430.30]	429.41
May 2027	434.36	[433.54, 435.17]	[433.11, 435.60]	434.47
Dec 2027	431.80	[430.72, 432.89]	[430.14, 433.46]	431.85
May 2028	436.77	[435.51, 438.02]	[434.85, 438.69]	436.90
Dec 2028	434.21	[432.72, 435.70]	[431.93, 436.49]	434.28
May 2029	439.18	[437.53, 440.82]	[436.66, 441.70]	439.34
Dec 2029	436.62	[434.75, 438.49]	[433.76, 439.49]	436.72
May 2030	441.59	[439.56, 443.61]	[438.49, 444.69]	441.78
Dec 2030	439.03	[436.78, 441.28]	[435.59, 442.47]	439.16